

My introduction

01 | Personal Information



- **Full Name:** Trinh Ngoc Huynh
- **Date of Birth:** 05/04/2002
- **Email:** huynhtn@vnu.edu.vn
- **LinkedIn:** <https://www.linkedin.com/in/huynh-spm/>
- **Github:** <https://github.com/huynhspm>
- Teacher assistant at the
Institute for Artificial Intelligence, University of Engineering and Technology

02 | Learning Path



- 2017 - 2020: Graduated from High School as the valedictorian
 - Major: Mathematics and Informatics
 - High School for Gifted Students (HSGS)
- 2020 - 2024: Graduated as the salutatorian with a Bachelor's Degree
 - Major: Information Technology
 - VNU University of Engineering and Technology (UET-VNU)
- 06/2024 - Now: Study master's program at UET-VNU
 - Major: Computer Science
 - Expected Graduation: June 2026
 - Preparing a plan to seek PhD opportunities at VUB, Belgium

03 | My Research

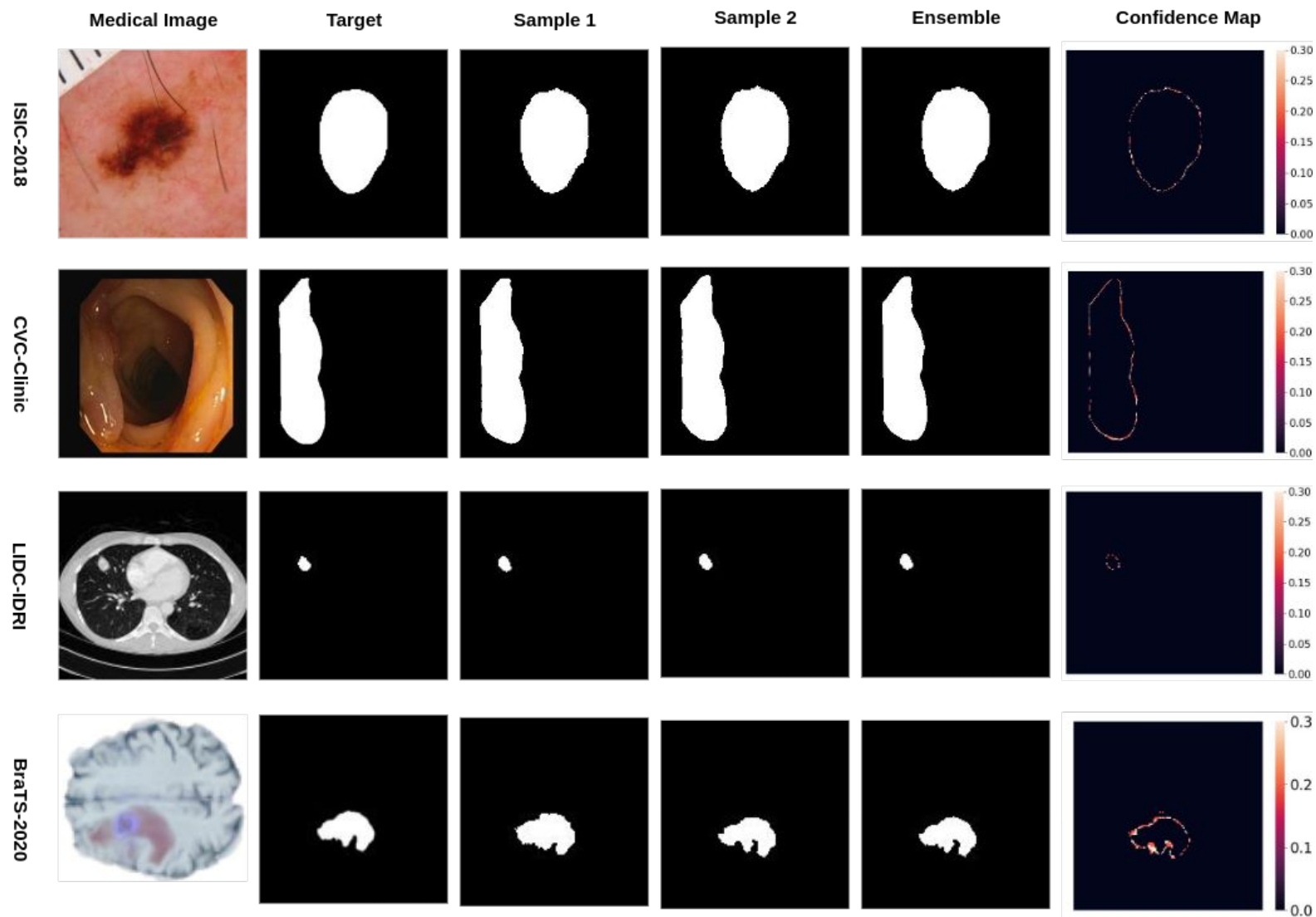


- Research Topic: Generative AI for Medical Image
- Key Projects:
 - **Latent Diffusion Models for Medical Image Segmentation**
 - **Aleatoric Uncertainty Medical Image Segmentation Estimation via Flow Matching**
 - **Lung Cancer Diagnosis through Chest CT Image Analysis**
 - **Latent representation learning for MS prediction**

03 | My Research

Latent Diffusion Models for Medical Image Segmentation

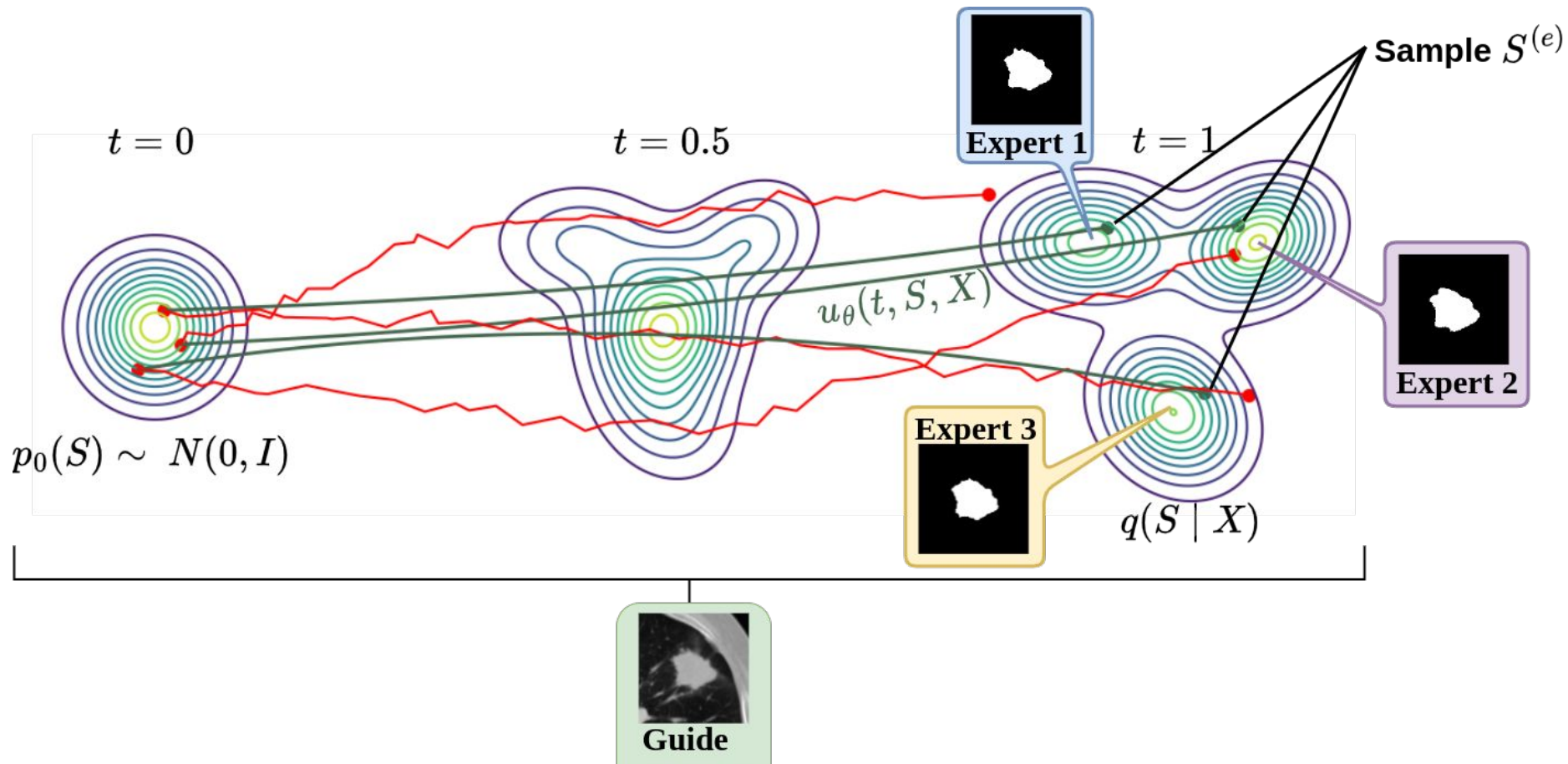
- Bachelor Thesis
- Latent Diffusion Models
- Ambiguity in Medical Imaging
- Multiple Outputs for Segmentation
- Ensemble Methods for Stable Prediction



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Aleatoric Uncertainty Medical Image Segmentation Estimation via Flow Matching

- Conditional flow matching
- Modeling multi-expert segmentation variability
- Multiple segmentation outputs
- Uncertainty map



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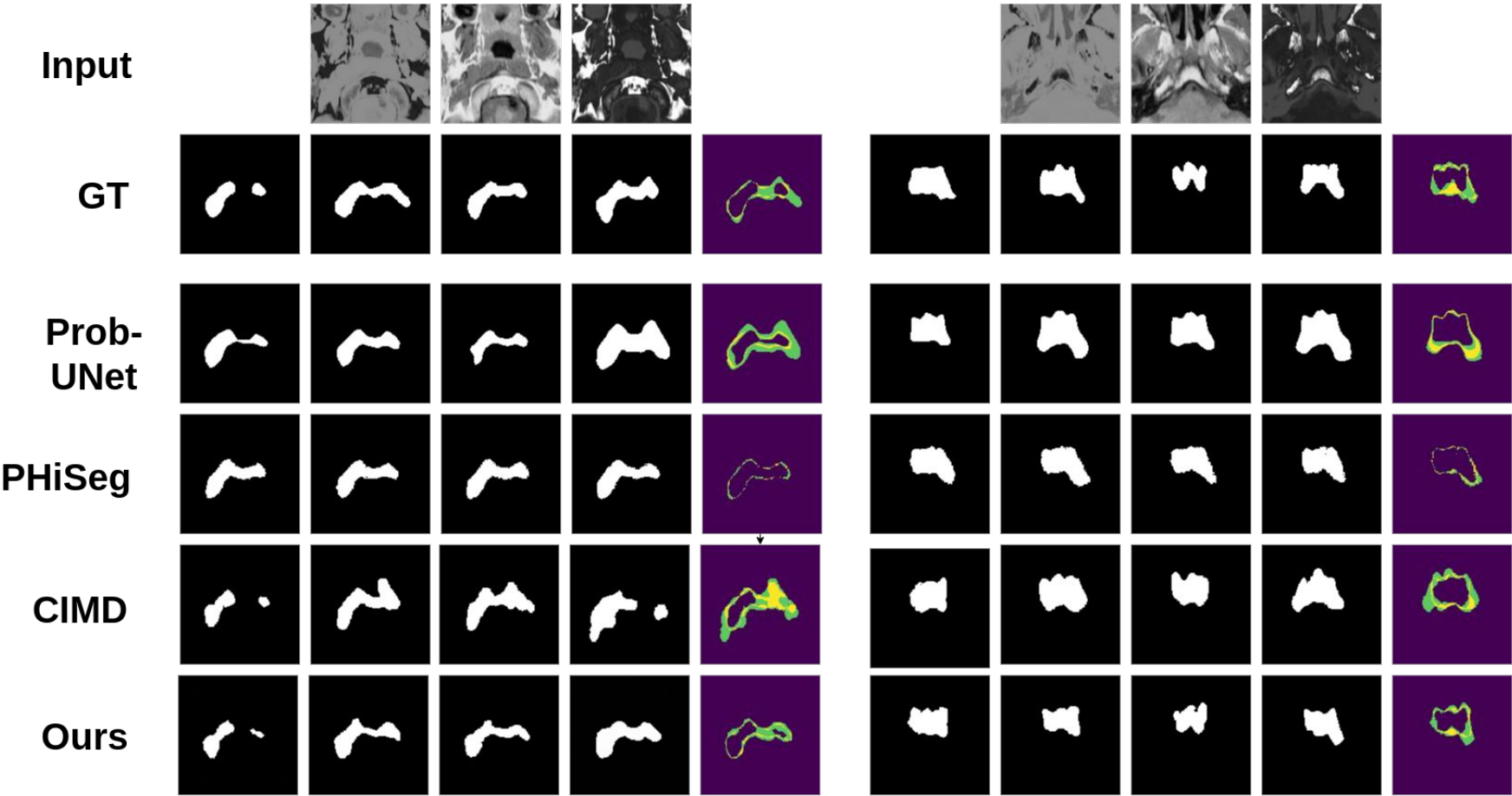
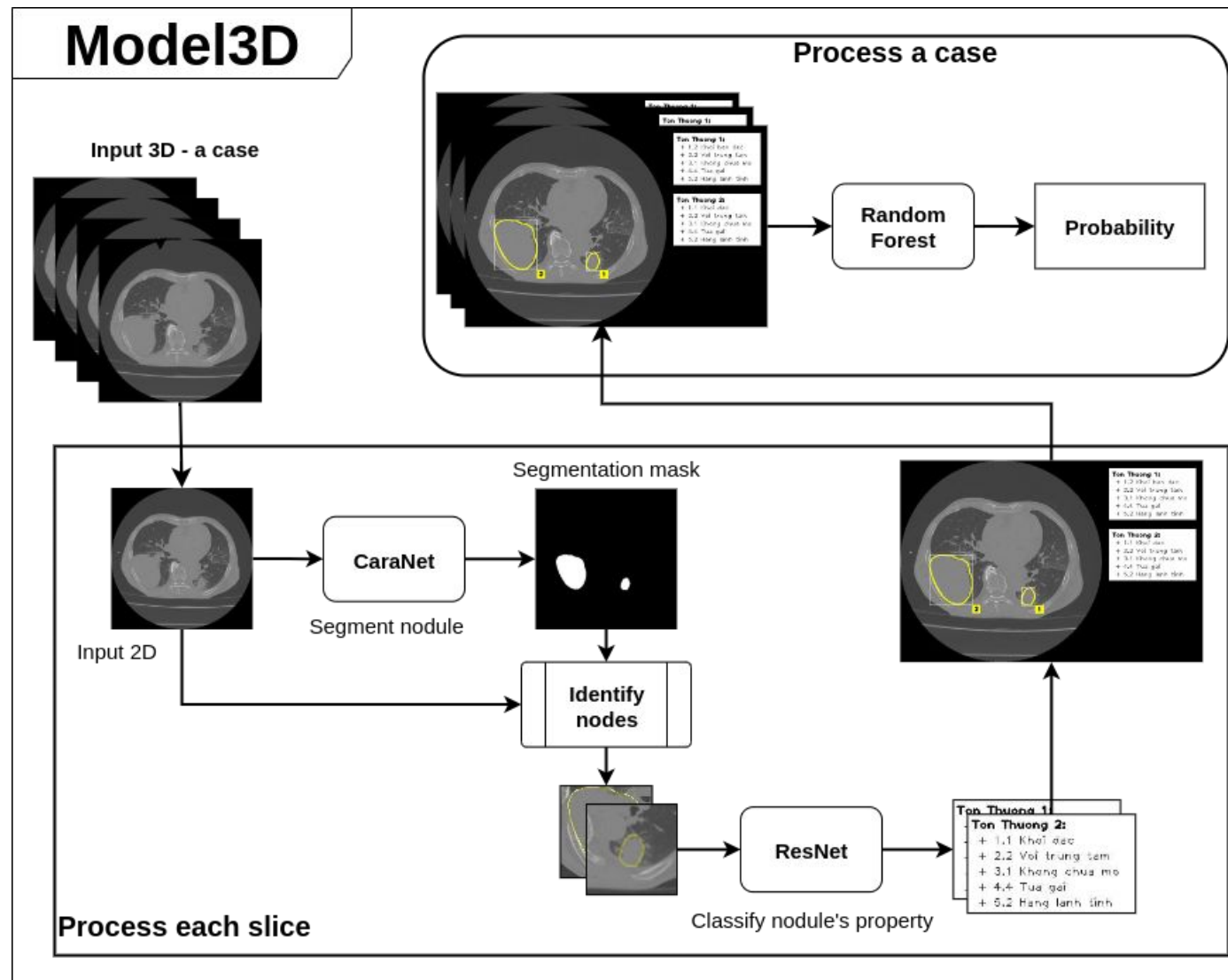


Figure 1. Qualitative comparison on MMIS: input images, followed by segmentation masks and uncertainty maps for GT, baselines, and our method.

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Lung Cancer Diagnosis through Chest CT Image Analysis

- Segmentation for each slice
- Identify nodules
- Classification for each nodule
- Aggregation
- Prediction for a case study

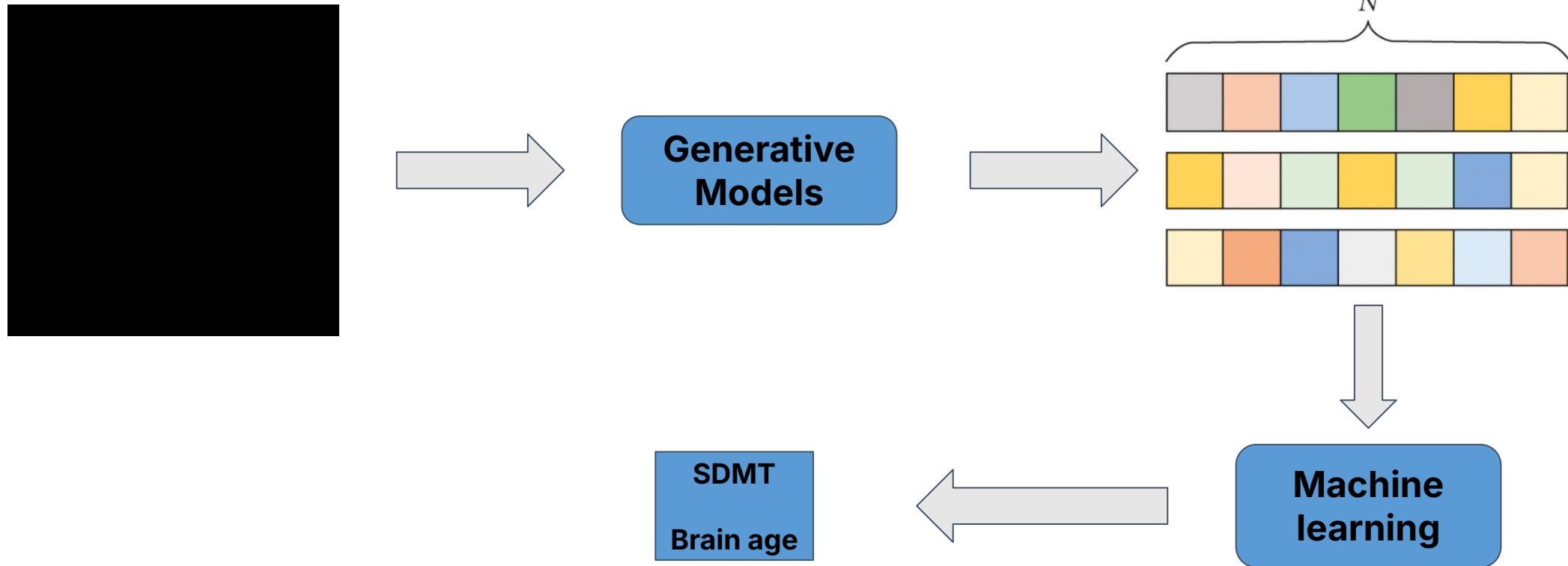


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Latent representation learning for MS prediction

Project UET–VUB:

- Assoc. Prof. Dr. Nguyễn Linh Trung
- Dr. Trần Quốc Long
- BSc. Nguyễn Đức Kiên
- BSc. Trịnh Ngọc Huỳnh
- Prof. Dr. MD. Guy Nagels

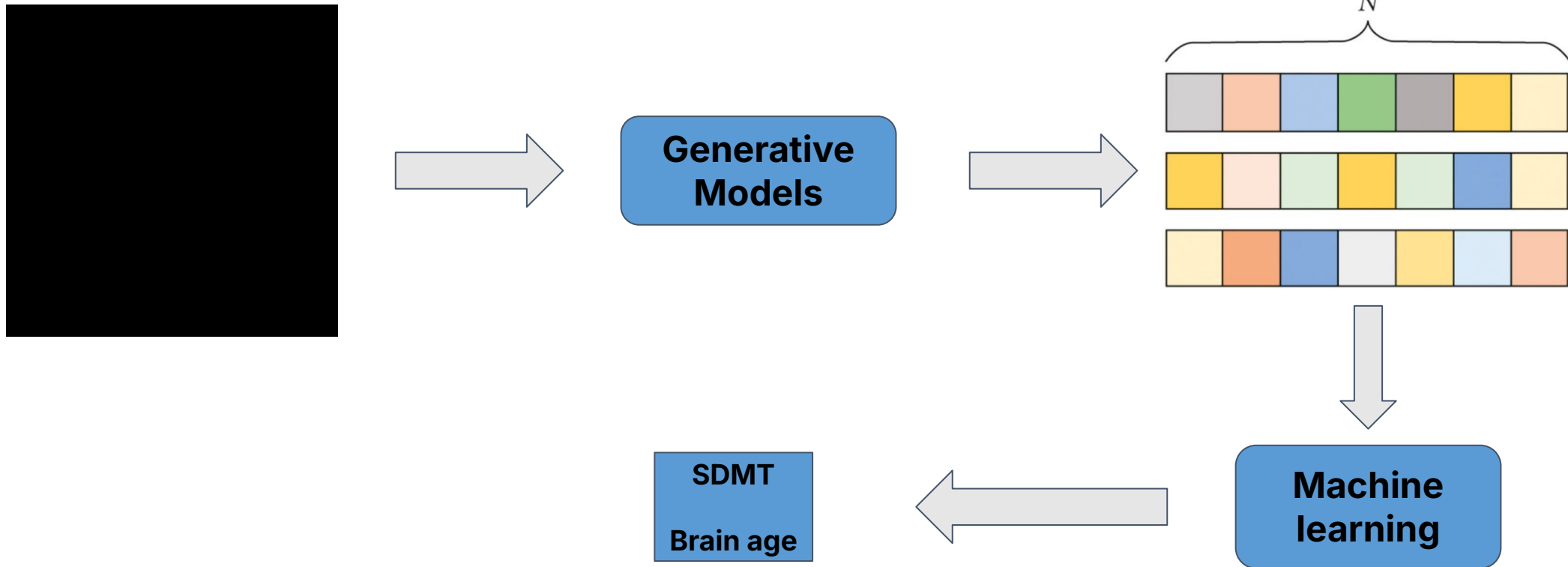


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Latent representation learning for MS prediction

Publication:

- VAE:
 - Paper submitted in *Machine Vision and Applications*
 - under review
- GAN: Kien, in the future
- Diffusion Model: Huynh, in the future



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Latent representation learning for MS prediction

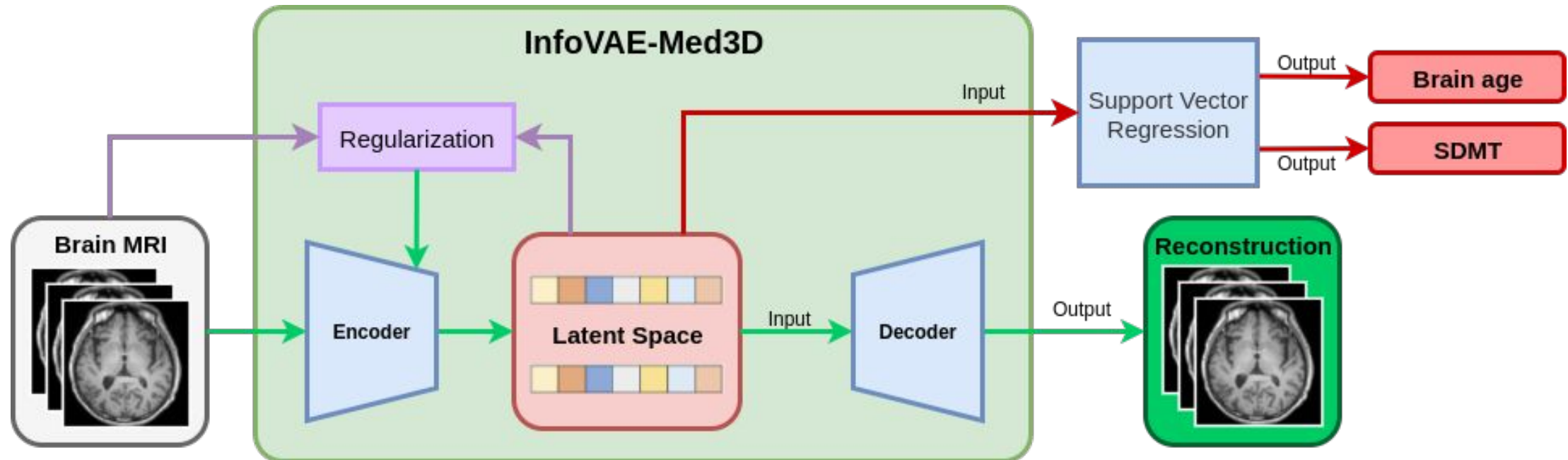
- [Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in Multiple Sclerosis](#)
 - Embedding Model: InfoVAE-Med3D
 - Regression Model: Support Vector Regression

BrainAge

- healthy control
- chronological age

Prague

- MS patient
- chronological age + SDMT



Latent representation learning for MS prediction

- Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in Multiple Sclerosis

BrainAge

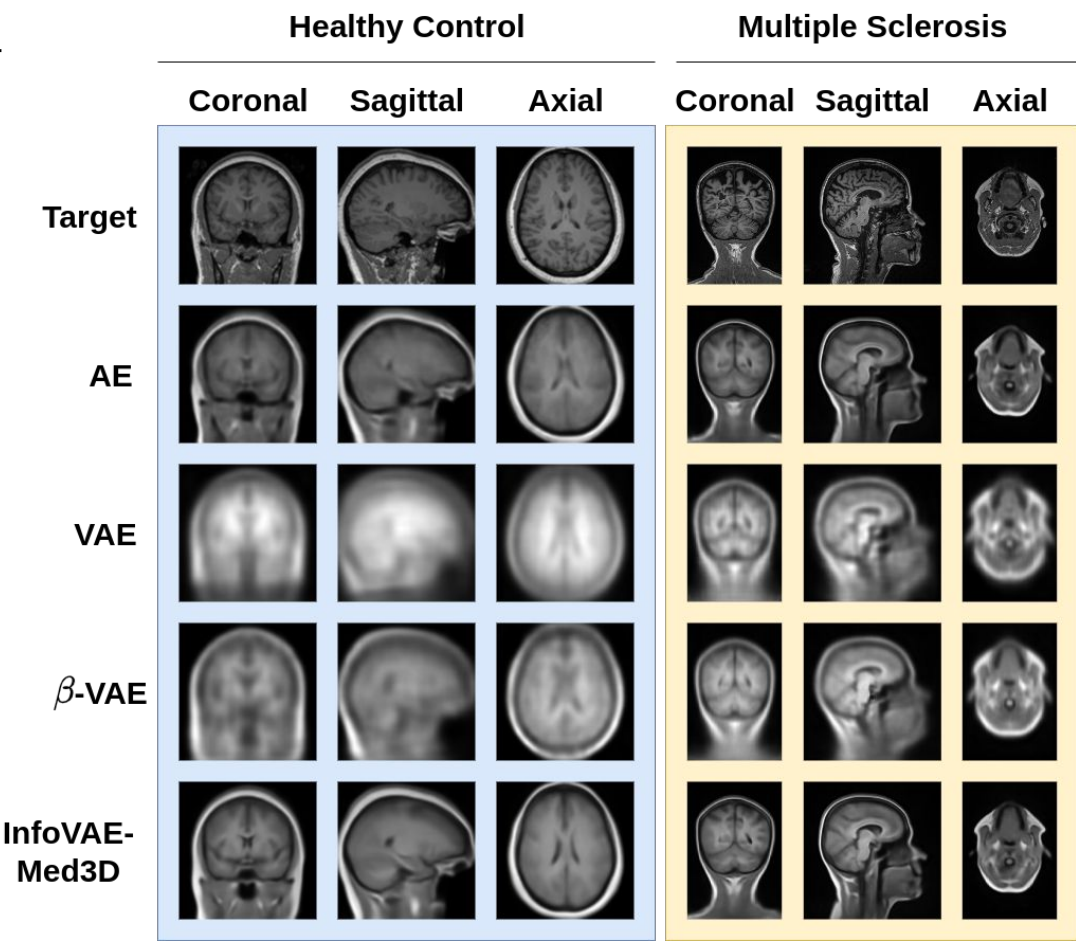
- healthy control
- chronological age

Prague

- MS patient
- chronological age + SDMT

Table 1: Reconstruction results on two datasets: our proposed model with multiple configurations compared against three VAE variants. The best results are highlighted in bold.

Models	BrainAge		Prague	
	SSIM	PSNR	SSIM	PSNR
AE	0.730	23.93	0.768	24.98
VAE	0.377	19.00	0.616	21.46
β -VAE	0.535	21.17	0.653	23.12
InfoVAE-Med3D ($\alpha = 1, \beta = 1$)	0.519	20.90	0.623	22.97
InfoVAE-Med3D ($\alpha = 0.001, \beta = 1$)	0.554	22.08	0.663	23.65
InfoVAE-Med3D ($\alpha = 0, \beta = 0.1$)	0.741	24.71	0.765	24.97
InfoVAE-Med3D ($\alpha = 0, \beta = 1$)	0.750	24.91	0.789	25.64
InfoVAE-Med3D ($\alpha = 0, \beta = 10$)	0.745	24.76	0.779	25.23



03 | My Research

Latent representation learning for MS prediction

- [Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in Multiple Sclerosis](#)
 - Downstream tasks:
 - Brain age prediction
 - SDMT prediction

Table 2: Quantitative results of downstream tasks on two datasets, comparing the proposed InfoVAE-Med3D with three VAE variants. The best results are highlighted in **bold**.

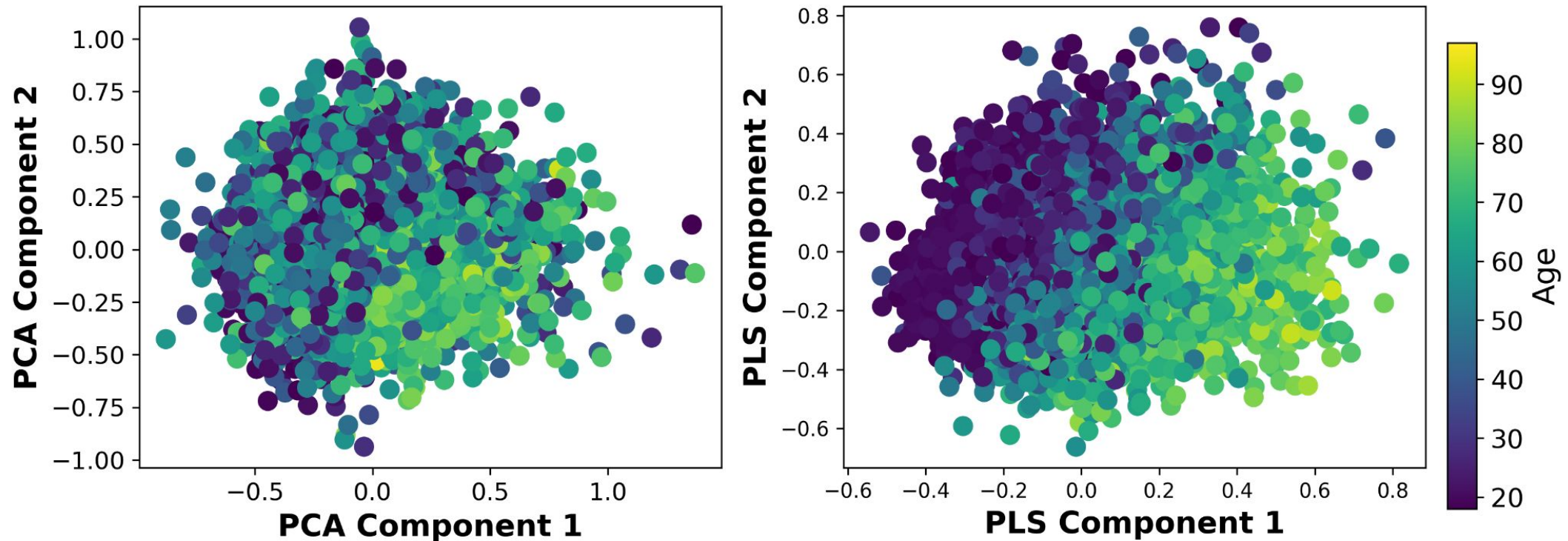
Models	BrainAge			Prague					
	brain age			brain age			SDMT		
	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓
AE	8.348	0.751	10.70	5.233	0.517	6.540	9.005	0.121	11.831
VAE	11.5675	0.425	16.249	7.717	-0.036	9.807	9.711	-0.018	12.35
β -VAE	9.957	0.648	12.820	7.652	-0.016	9.710	9.709	-0.017	12.33
InfoVAE-Med3D	7.664	0.778	10.09	4.736	0.639	5.947	8.531	0.160	10.833

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Latent representation learning for MS prediction

- [Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in Multiple Sclerosis](#)
 - Downstream tasks → Visualization for interpretation
 - Principal Component Analysis (PCA): unspecified method
 - Partial Least Squares regression (PLSRegression): supervised method

→ *Brain age in latent space shows a smooth transition across the 2D space from low to high values.*

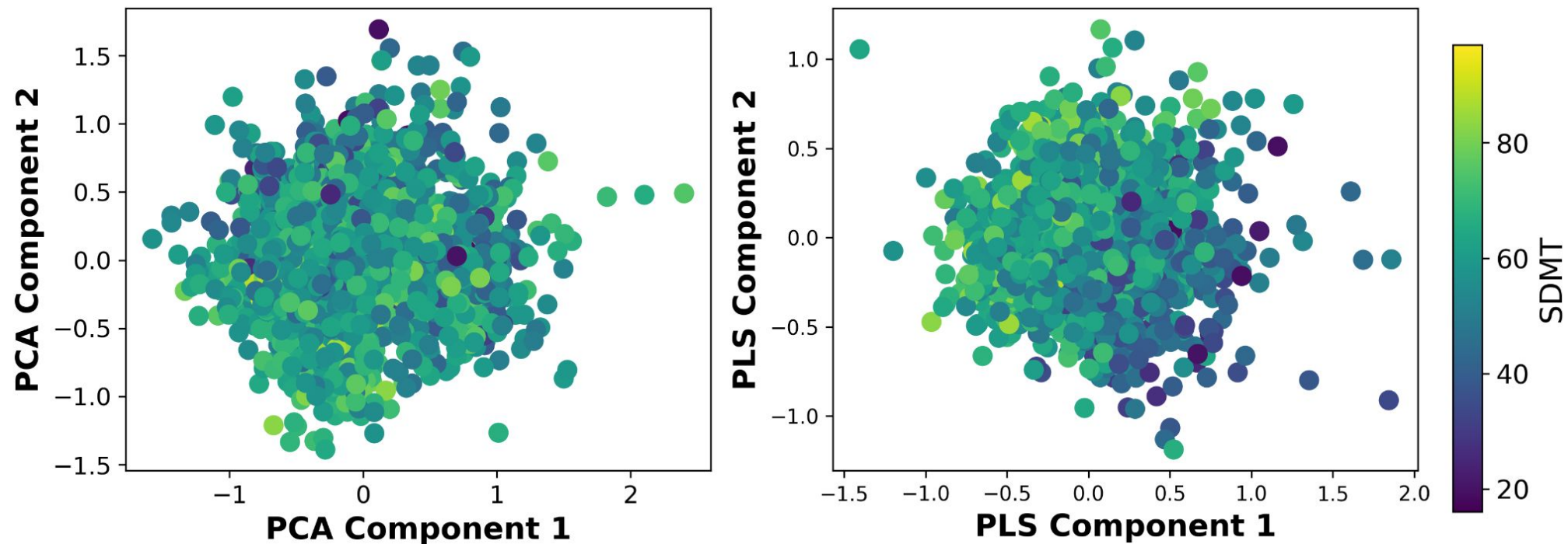


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→ → *Latent space exhibits smooth transition related to SDMT, but less distinct than age or gender*

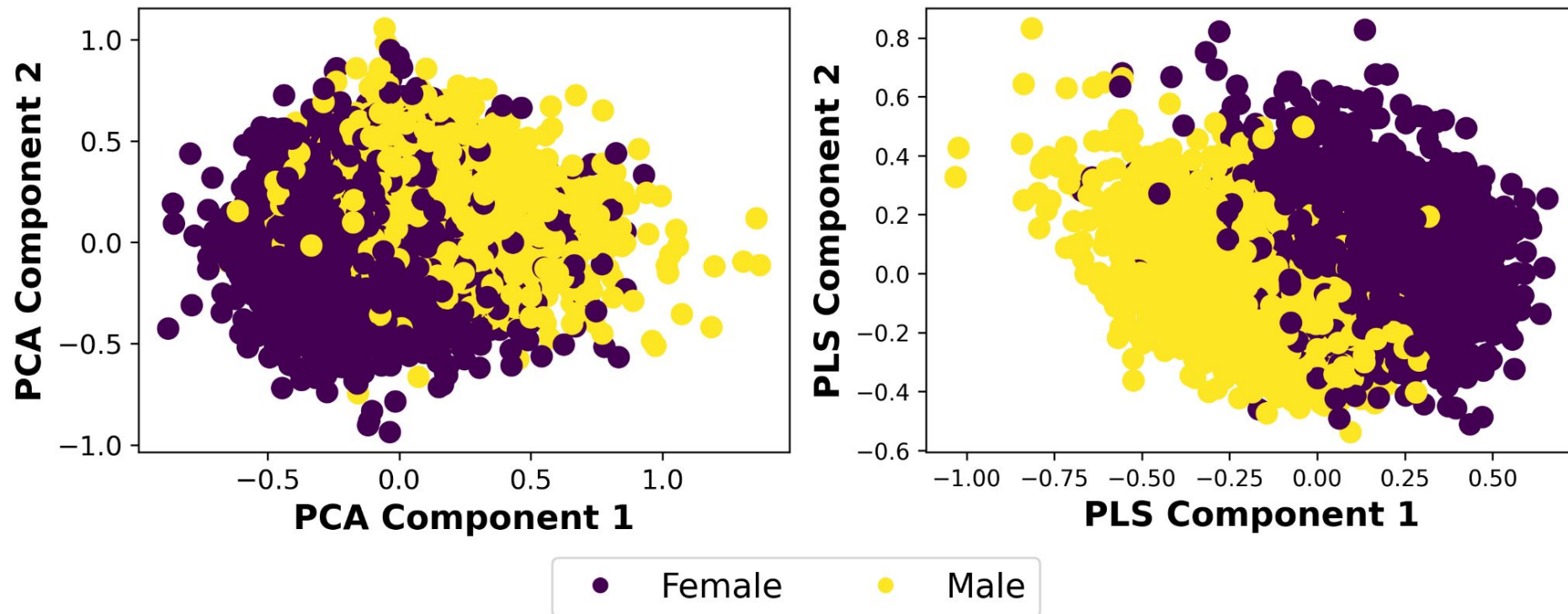


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→ → *two most variance dimension in latent space related to gender information*



Thanks For Listening
